MS&E 226: Fundamentals of Data Science
Project Part 1: Data Exploration and Prediction

Overview

In MS&E 226, you will work on a quarter-long project to give you a chance to apply some of the concepts from class in the context of a real dataset. This handout gives details on the first part of the project.

- This part of the project is due on **November 4, 2026 at 11:59 PM PT** (submission on Gradescope).
- Each part below comes with a checkbox. This is what you should submit to receive credit for that part.
- Each part of the project noted below will be graded using the following rubric:
  - You will receive zero points if you do not attempt it.
  - If you attempt it, but there are substantial methodological errors or major conceptual misunderstandings of the material, you will receive 2 points.
  - If you attempt it and there are no substantial methodological errors or major conceptual misunderstandings in what you submit, you will receive 3 points.

  We expect students who make a reasonable effort, even if not perfect, will receive 3 points. Receiving 2 points is intended to be a sign of significant comprehension issues.
- You may choose to use more text for some responses, and less text for others. Your entire report (when compiled to PDF) should be no longer than **5 pages, no less than single spaced, 11 point Times or equivalent font, with no smaller than 1 inch margins**; you may include up to **one additional page for figures only**. If you submit material beyond the requirements, the TAs will only grade the first 5 pages of text (and one page of figures). **Upload your code as a zip file to a cloud service and include a download link in your report (see Section 9).**

1. Stakeholder presentation preparation

At the end of the quarter, you will give an oral presentation about your project as if you are presenting to a stakeholder who cares about the outcome of your data analysis. This stakeholder could be a business leader, policy maker, nonprofit director, or other decision-maker who would use your findings to inform real-world choices. A major goal of your work in the written reports is to prepare you for this presentation.

NOTE: Your presentation will be evaluated on motivation, evidence, recommendations, and technical literacy. Every team member should be able to explain and justify the methods, features, tuning, and evaluation used, including their objectives, assumptions, and limitations. This applies equally to work developed with AI assistance.

☐ (a) Identify a realistic stakeholder for your dataset and analysis. Who would care about the relationships you're exploring? Briefly describe this stakeholder (e.g., their role, organization type, and why they would commission this analysis).

☐ (b) What specific questions or decisions would your predictive models help this stakeholder answer or make? Specify when a prediction would be made and what information would be available at that moment.

2. Outcome

☐ Choose a continuous or binary variable you plan to use as your outcome variable. Provide a brief explanation of your choice, including how this variable relates to your stakeholder's interests. (*Note:* You can always create a binary response variable by starting with a continuous variable Y, defining a binary variable Z that is one if Y is larger than a threshold, and zero otherwise. If you choose this approach, describe how you chose the threshold; note the threshold **should not** be a function of your data.)

3. Dataset

Your selected dataset must satisfy the following requirements:

(i) It should have at least a few thousand rows and at least 10 predictor columns that you intend to use for the project, excluding the outcome and extraneous columns such as IDs.

(ii) It should have a variety of covariates, both numeric and categorical.

(iii) It should contain real-world data, not synthetically generated, algorithmically generated, or simulated data.

(iv) Choose a dataset whose rows can reasonably be treated as independent observations. Time series or panel datasets are not allowed, and time should not be included as a feature in your dataset.

Of course, all data is inherently collected over time; however, prediction and inference tasks involving time series bring additional complexity and need additional care and methodology that is beyond the scope of MS&E 226. (You should learn more about this, but after you've mastered the material in this class!)

Examples of time series or panel data include: financial time series data; sensor data from wearables such as heart rate monitors; weather time series; etc.

(v) Some missing data is ok, but excessive missing data is a red flag and such datasets should be avoided.

☐ (a) Confirm that your dataset meets the technical criteria (i)-(v) above. Identify what one row (i.e., one observation) represents. Include a link/citation/acknowledgment to its source/creator, and acknowledge your permission to use the data.

☐ (b) Briefly describe your chosen dataset, and why the particular choice of data sample, and the features available, make it appropriate for addressing your stakeholder's decision-making needs.
☐ (c) Before continuing, remove any duplicate records of the same observation (and confirm you have done so).

4. Holdout set

After checking eligibility and removing duplicate records, and before exploratory analysis, preprocessing, or fitting models, randomly set aside 20% of the observations as a holdout. Use a fixed random seed and save the split. The remaining 80% is your **development data**: use it for all exploration, preprocessing, and modeling in Part 1. Keep the holdout untouched until Part 2, including withholding it from AI tools and agents.
☐ Acknowledge that you have set aside 20% of your data, report the split seed and sample sizes, and describe how you kept the holdout data out of your own and any AI-assisted development.

5. Data description and exploration

☐ (a) Describe the process that led to the data being collected. Describe any issues with the data collection process (e.g., accuracy, completeness, sampling strategy, etc.) that could impact your subsequent data analysis.
☐ (b) Which covariates are most strongly correlated with your outcome variable? Which ones are most weakly correlated with your outcome variable? Describe any correlations that are surprising or unexpected to you. For categorical covariates, compare outcome distributions across categories. Use appropriate plots to illustrate these relationships.
☐ (c) Among those covariates identified in the last part, which covariates are most strongly correlated with each other (whether positively or negatively)? Again, describe any correlations that are surprising or unexpected to you. (For relationships involving categorical covariates, you may use comparisons across categories or contingency tables, as appropriate.)
☐ (d) Are there any natural subgroups in your data that you think will be important to your analysis (e.g., categorical variables that encode grouping)? Briefly describe any interesting quantitative differences between these groups (e.g., by comparing means of covariates across the groups).
☐ (e) Are any values in your dataset missing, e.g., `NULL`, `NA`, or blank values? Briefly describe how you will deal with missing values: do you plan to remove columns with missing values, or impute the missing values? Explain why you chose the approach you used.

6. Prediction

In this section you build and compare predictive models.
☐ (a) Describe the metric(s) you will use to measure prediction error, identify the primary metric used for model selection, and explain how it relates to your stakeholder's decision. Be precise in your answer.
☐ (b) Describe how you will train, tune, and compare models using only the development data. Specify your validation split or number of cross-validation folds, random seeds, and fitting procedure. Ensure you use the same evaluation splits for all comparisons.

Fit any data-dependent preprocessing, such as imputation, scaling, or automated feature selection, using only the training portion of each split. Apply the fitted preprocessing to the corresponding validation portion.

☐ (c) Choose a simple baseline appropriate to your metric, justify it, and report its performance using the evaluation procedure in part (b). Examples include predicting the training-sample mean for a continuous outcome, or the training-sample class proportions for binary outcome. Explain what comparison to this baseline tells your stakeholder.
☐ (d) **This part is graded out of six points.** Specify your learning algorithms and feature sets:

- Choose at least three distinct learning algorithms. Include linear regression for a continuous outcome or logistic regression for a binary outcome, and at least one method outside the linear-model family, such as a decision tree or random forest. (Note: Linear and logistic regression, ridge, and lasso are all linear models.) Briefly explain why you chose each algorithm and how it produces predictions. The baseline does not count toward the three algorithms, and changing only features or tuning parameters does not count as a different algorithm.

- Define at least two feature sets to use with every algorithm. Explain the rationale for their differences and why those differences might affect prediction. You may add engineered features, transform existing predictors, or remove a group of predictors. You may use a separate feature-selection procedure, though this is not required as long as your choices are well-motivated. You may find it useful to apply exploratory findings from Section 5 to motivate your feature sets.

Emphasize careful evaluation and clear explanations; more complex or more numerous models, features, and experiments do not necessarily make for a stronger project. You may use AI to engineer features, implement models, and automate experiments.
☐ (e) Train and evaluate every algorithm/feature-set combination using the procedure in part (b). Present one table comparing their validation performance and the baseline. Report the tuning settings tried and selected for each combination, including the scope of any automated search (e.g., number of trials or time budget). Which algorithms perform better for a given feature set? How does changing the feature set affect each algorithm's performance? Discuss whether the observed differences are practically meaningful to your stakeholder.
☐ (f) Select your final algorithm/feature-set combination using your primary metric and explain and interpret your choice.

Save your selected model and all steps needed to generate predictions, including fitted preprocessing, feature definitions, tuning choices, and any classification threshold. You may refit on all development data before saving. You will evaluate this saved pipeline on the untouched holdout in Part 2.

7. Stakeholder Guidance

☐ (a) What decisions, if any, does your analysis support for your stakeholder? Explain your model's performance in terms they would understand, and identify further evidence needed before acting.
☐ (b) What are the practical limitations of your predictive models that your stakeholder should be aware of? Discuss factors like prediction accuracy, potential biases, situations where the models might not be reliable, or conflation of correlation with causation.

8. AI tools

☐ List any AI tools you used, including model/version information where available. Briefly describe the tasks you delegated and how you checked the resulting code, features, or analysis. If you used AI, give one concrete example of an important suggestion you accepted, modified, or rejected and

explain your decision. If you used no AI tools, state that. You should not submit chat transcripts or raw AI output.

9. Code upload

☐ Using Stanford's Google Drive or a similar cloud service, upload your code as a zip file, make sure it is viewable/downloadable by the teaching staff, and paste a URL in your project report that links to this zip file. Include the code needed to reproduce the data split, preprocessing, feature construction, model selection, and final fitted pipeline, together with random seeds, package versions, and data-access instructions. Do not include data you lack permission to redistribute.